

BLOODSTOCK AI

HOW TO ANALYSE A BREEZE-UP HORSE WITH EFFICIENCY AND PRECISION

This report presents a comprehensive framework for evaluating thoroughbreds in breeze-up sales, combining traditional bloodstock expertise with the cutting-edge analytical capabilities of BloodstockAI.

From biomechanical metrics to hoof conformation analysis, discover how data-driven insights are transforming how buyers identify elite racehorses before the hammer falls.

80%

of career-ending issues
linked to hoof problems

14.25

m/s — top speed achieved
through biomechanics

100%

objective analysis vs.
human intuition alone



The Challenge of Evaluating Horses in Breeze-Up Sales

Breeze-up sales represent one of the most exciting and high-stakes moments in the thoroughbred market. In a matter of seconds, a horse gallops past a crowd of buyers, agents, and trainers — each attempting to assess years of potential racing ability from a fleeting glimpse. The decisions made in this environment involve millions of dollars and the careers of horses, jockeys, and connections alike.

Traditional evaluation methods rely heavily on the experienced human eye: a skilled bloodstock agent reads the horse's movement, conformation, and attitude based on instinct built over decades. This 'clinical eye' is invaluable — but it has limits. Subjectivity, fatigue, incomplete information, and the sheer speed of the breeze itself all conspire against consistent, objective assessment.

The question is no longer whether data analysis can contribute to bloodstock evaluation — it already does. The question is how to integrate it most effectively alongside traditional expertise.

2–3 sec Time a buyer has to observe a breeze	£500K+ Average price of a top breeze-up lot	80% Career issues traced to hoof/limb problems	14.25 m/s Top speed calculated from biomechanics
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This report outlines the core pillars of efficient breeze-up horse analysis and explains how **BloodstockAI** brings precision, objectivity, and data-driven intelligence to each of these pillars.



Why Two Horses with the Same Time Are Not Equal

The fundamental insight of modern performance analysis is deceptively simple: a stopwatch tells you the outcome, not the story. When two horses post identical breeze times, the conventional reading is that they are equivalent. Biomechanical analysis reveals a far more nuanced reality.

The Core Equation

Locomotion speed in thoroughbreds is governed by two mechanical variables:

Stride Length × Stride Frequency = Speed

6.2 m × 138 strides/min = 855 m/min

855 m/min ÷ 60 = 14.25 m/s

14.25 × 3.6 = 51.3 km/h

Official timers measure the result of this equation. They do not reveal the mechanical composition that produced it. No one calculates this visually on a platform in real time. Frame-by-frame analysis is the only reliable method to decompose the equation.

The Two-Horse Paradox

Horse A	Metric	Horse B
6.8 m	Stride Length	5.6 m
126/min	Stride Frequency	153/min
More efficient, more reserve	Assessment	Working harder, tires sooner
Higher ceiling	Long-term Potential	Fatigue risk

Both horses posted identical breeze times. But Horse A achieves that speed with greater mechanical efficiency — a longer stride means each footfall generates more forward momentum per unit of effort. Horse B is already operating near its mechanical ceiling. Only frame-by-frame analysis can reveal this distinction. BloodstockAI performs this decomposition automatically, flagging efficiency ratios and fatigue risk indices for every horse in the sale catalogue.



Combining the Clinical Eye with Artificial Intelligence

The best bloodstock agents in the world rely on their 'clinical eye' to evaluate a horse's conformation. Years of experience allow them to spot subtle details in structure, balance, and movement — the slight offset of a knee, the angle of a pastern, the breadth of a chest. These observations, accumulated over thousands of hours at sales and racetracks, represent an irreplaceable form of expertise.

But what if that same process could be enhanced with precise, quantitative data analysis?

Instead of relying solely on human intuition, BloodstockAI combines:

- **Biomechanical Analysis** — Stride length, frequency, symmetry, ground contact time, and energy transfer efficiency
- **Conformation Measurements** — Structural geometry assessed from standardised photography and video
- **Pedigree Data** — Cross-referenced genetic lines correlated with performance outcomes on specific going and distances
- **Performance History** — Workload data from pre-sale training logs and timed gallops
- **Movement Pattern Recognition** — AI-trained models identifying gait abnormalities invisible to the naked eye

The Future of Bloodstock Analysis

All five data streams are processed together by BloodstockAI to generate objective, reproducible insights — available instantly, before the auction begins.

This is not a replacement for the expert eye. It is a force multiplier for it.

Key Conformation Checkpoints at Breeze-Up

Point	Ideal Range	Clinical Significance
Shoulder Angle	45–55° ideal	Directly impacts stride length and ground-covering ability
Pastern Slope	45–50°	Shock absorption; deviations predict tendon strain
Cannon Bone Length	Shorter = better	Lever mechanics; shorter cannons reduce injury risk
Hip Height vs. Withers Height	Hip ≥ Withers	Growth potential indicator in young stock
Chest Width	Wide but not exaggerated	Lung and heart capacity; airway efficiency
Hindquarter Muscling	Deep gluteal development	Propulsive power; acceleration capacity



The Most Overlooked Dimension of Breeze-Up Evaluation

The majority of buyers at breeze-up sales focus their attention almost exclusively on the gallop and the pedigree. Conformation is assessed quickly, often from a distance. The hooves — arguably the single most important structural feature for long-term soundness — are frequently ignored entirely. This is a critical oversight with measurable consequences.

80%

of all career-ending conditions in thoroughbreds are directly linked to limb and hoof pathology. This is not a marginal statistic. It means that hoof evaluation at point of purchase is arguably more predictive of long-term racing career than pedigree alone.

Critical Hoof Pathologies and Their Long-Term Consequences

◆ Long Toe / Low Heel (LTLH)

The single most common structural fault in thoroughbred yearlings and two-year-olds. LTLH places catastrophic mechanical stress on the deep digital flexor tendon. Horses with this presentation **inevitably develop tendinitis** under training loads. The fault is detectable in a lateral hoof photograph — and BloodstockAI identifies it automatically.

◆ Contracted Heels

Narrow, pinched heel bulbs restrict blood flow and shock absorption. Horses with contracted heels are on a predictable trajectory toward **navicular syndrome within 3–4 years** of hard training. This is rarely flagged by buyers at breeze-up sales, yet it is visible to any experienced farrier — and now to BloodstockAI.

◆ Hoof Pastern Axis Misalignment

Broken-back or broken-forward hoof-pastern axis creates uneven load distribution across the coffin joint, navicular bone, and associated ligaments. Chronic lameness follows without aggressive corrective farriery — which itself adds cost and interrupts training.

◆ Medial/Lateral Imbalance

Unequal medial-lateral hoof wall height causes rotational loading through the fetlock and knee. Often the root cause of splints, osselets, and chronic low-grade lameness that manifests only after sustained training.

BloodstockAI Hoof Intelligence

An experienced farrier can identify LTLH, contracted heels, and axis misalignment from a single lateral photograph. BloodstockAI applies the same diagnostic framework at scale — analysing every horse in the catalogue, flagging risk profiles, and ranking hoof health scores alongside biomechanical and conformation data.

The buyer using BloodstockAI avoids purchasing problems that other buyers cannot see.



Precision Intelligence for the Modern Bloodstock Market

BloodstockAI is purpose-built for the bloodstock professional who understands that edge in the auction ring is not found in information that everyone has — it is found in the analysis that others cannot perform. The platform synthesises multiple analytical streams into a single, actionable intelligence layer overlaid on every horse in the sale.

Platform Capabilities

- 01

Frame-by-Frame Video Analysis
Automated decomposition of breeze footage into stride-by-stride biomechanical data. Stride length, frequency, symmetry index, ground contact time, and propulsion angle are calculated for every stride of every horse.
- 02

Conformation Scoring Engine
Standardised photography processed through trained computer vision models. Structural geometry is measured, scored, and benchmarked against a database of thoroughbreds with known performance outcomes.
- 03

Hoof Pathology Detection
Lateral and dorsal hoof images assessed for LTLH, contracted heels, axis deviation, and balance asymmetry. Risk scores are generated with farrier-grade diagnostic precision.
- 04

Pedigree × Performance Correlation
Bloodline data cross-referenced with going preference, distance aptitude, and historical performance of close relatives — weighted by generation proximity.
- 05

Integrated Risk Profile
All data streams combined into a single horse-level risk/opportunity score, enabling rapid prioritisation of targets before and during the sale.

The Competitive Advantage

In a market where every serious buyer has access to the same pedigree data, the same official breeze times, and often the same veterinary inspection reports, marginal advantage lies in the quality of analytical interpretation. BloodstockAI does not replace the expert — it equips the expert with capabilities that no competitor without the platform can match.

Without BloodstockAI	With BloodstockAI
Stride length estimated by eye	Stride length measured frame-by-frame
Hoof structure overlooked at speed	Hoof pathology flagged before you arrive
Two horses with same time = equivalent	Mechanical efficiency ratio distinguishes them
Pedigree assessed from memory/gut	Pedigree correlated with quantified outcomes
Risk discovered post-purchase	Risk quantified pre-purchase



The Future of Breeze-Up Analysis

The breeze-up market rewards buyers who see what others miss. For generations, that meant cultivating an exceptional human eye — and the best agents still possess this in abundance. But the ceiling of the human eye, operating in real time under auction pressure, is finite.

BloodstockAI exists at the intersection of traditional bloodstock expertise and the analytical power of modern machine intelligence. It does not attempt to replace judgment — it provides the objective data upon which better judgment can be made.

◆ **Biomechanics**

Identical times can hide radically different mechanical efficiency profiles. Only frame-by-frame analysis reveals the horse behind the stopwatch.

◆ **Conformation**

AI-measured structural geometry provides reproducible, benchmarkable data that complements the expert eye without replacing it.

◆ **Hoof Analysis**

The most neglected dimension of breeze-up evaluation is also the most predictive of long-term soundness. BloodstockAI makes it a standard part of the process.

◆ **Integration**

When biomechanics, conformation, hoof health, pedigree, and performance data are combined, the resulting intelligence profile is qualitatively superior to any single data stream.

BloodstockAI — The Intelligent Edge

The buyer who uses BloodstockAI enters every sale with information their competitors do not have. They avoid the horses others will regret buying.

They identify the horses others will regret not buying.

This is not the future of bloodstock analysis. It is the present.

For professional enquiries and platform access, contact BloodstockAI.